

Announcements

- ▶ Grading: I will catch up soon. I'm sorry!
- ▶ Happy (perfect square)³ day! (and nice pythagoras day!)
(in this perfect-square year!)
- ▶ Schedule:
 - Today (Lecture)**: Electromagnetic Spectrum (light) and Blackbody radiation
 - Tomorrow (Lab)**: “all the light you cannot see”
 - Thursday (Discussion)**: Blackbody radiation and estimating Earth's Temperature
- ▶ Next Week: Greenhouse Effect I

Today's Reading Quiz (3 min)

On your own sheet of paper. . .

How does the Sun's energy get to the Earth? How would you describe that energy/energy-transfer, and what are other examples (not the Sun, but same type of energy transfer)?

Turn in to Blackboard as scanned pdf (or image is fine).

Review: Energy Types, Energy Transfer, Power

Energy Types (units of Joules = "J")

- ▶ "External": Kinetic (K) of object, Potential (V) in gravitational or spring "field"
- ▶ "Internal": Thermal or mechanical kinetic/potential of atoms or internal parts; Chemical; Nuclear
- ▶ (new today) "Electromagnetic": carried by photons/light

Energy Transfer Mechanisms (units: J)

- ▶ Mechanical Work: $W = F \times \Delta x$ (force over a distance)
- ▶ Heat Conduction: Objects in contact (high T to low)
- ▶ Convection: fluid motion of internal parts
- ▶ (today) Radiation: transmission of photons

Power: Energy per time ($P = \frac{E}{\Delta t}$, in units of $\frac{J}{s}$)

Flux: Power per area ($F = \frac{E}{\Delta t \times A}$, in units of $\frac{J}{s \times m^2}$)

One last good day for the professor

Let's sketch an energy flow diagram for the professor climbing Mount Ventoux ($h = 1900$ m).

First consider the time interval of “the climb”

One last good day for the professor (part 2)

Now consider the time interval of “the climb” and “the previous day’s meals” ($2000 \text{ calories} = 2000 \text{ kcal}$; $1 \text{ kcal} = 4200 \text{ J}$)

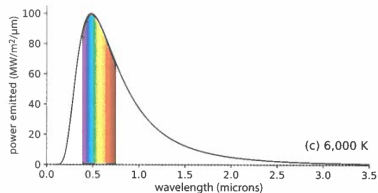
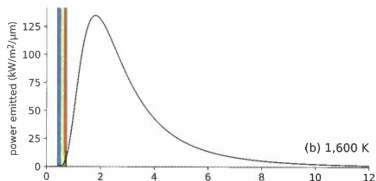
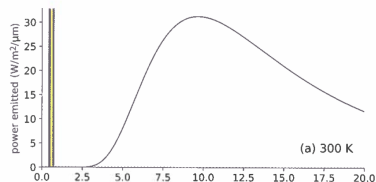
One last good day for the professor (part 3)

Q: Where is all of the food energy going?

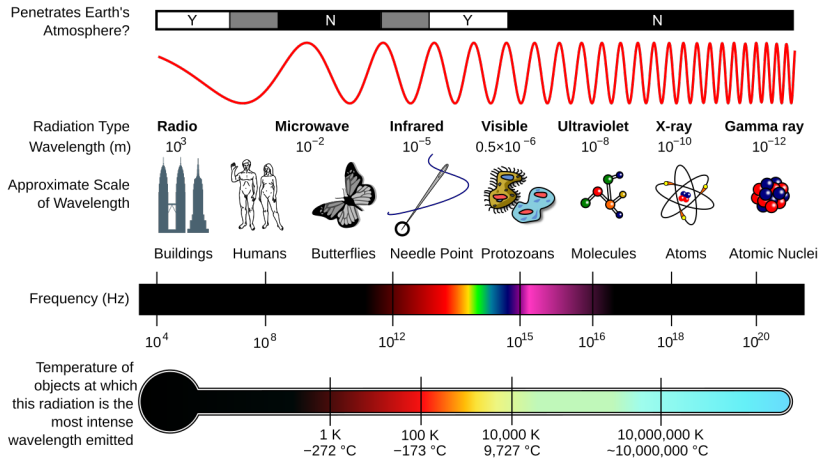
A: Turns out the professor is a light bulb.

Any object with a temperature (> 0 K) emits electromagnetic radiation

(There is also similar heat losses due to convection/conduction.)



Electromagnetic Radiation (light!)



Equations for Blackbody Radiation

Wien's Law

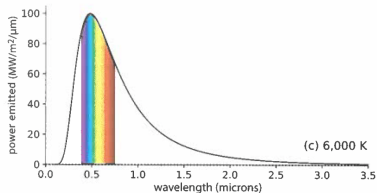
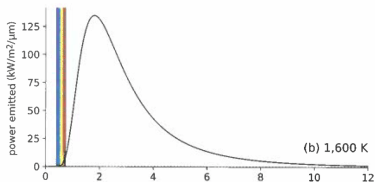
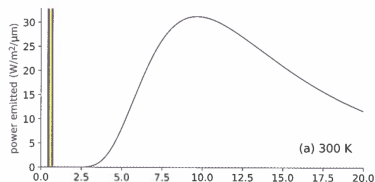
$$\lambda_{\text{peak}} = \frac{2900 \text{ } \mu\text{m} \cdot \text{K}}{T}$$

Stefan-Boltz. Law (total flux)

$$F = \left(6 \times 10^{-8} \frac{\text{J/s}}{\text{m}^2 \cdot \text{K}^4} \right) T^4$$

Peak "Spectral Flux"

$$F_{\lambda} = \frac{F}{1.5 \times \lambda_{\text{peak}}}$$



Graphing Bb Radiation

Let's graph (on the log-log paper provided) the Bb radiation of three objects:

- ▶ Sun ($T = 6000 \text{ K}$)
- ▶ Human ($T = 310 \text{ K}$)
- ▶ Liquid Nitrogen ($T = 77 \text{ K}$)

We use Wien's law and the "spectral flux" peak, as well as a 10% width of $(\lambda/2, \lambda \times 3)$.